

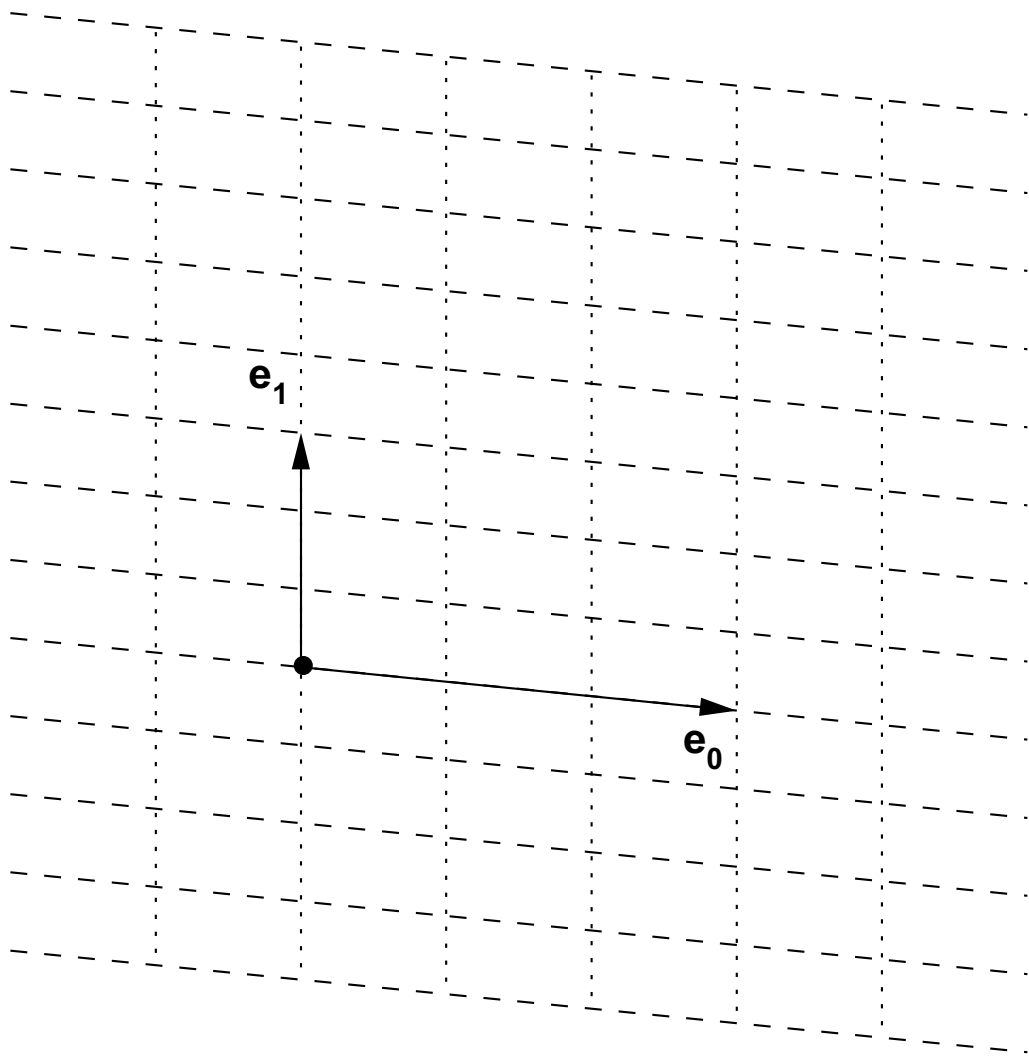


Figure 4.1 Let arrows $\mathbf{e}_0$ and $\mathbf{e}_1$ depict the vectors of a basis vector field at a particular point. Then the foliations shown by the parallel lines depict the dual basis one-form fields at that point. The dotted lines represent the field $\tilde{\mathbf{e}}^0$ and the dashed lines represent the field $\tilde{\mathbf{e}}^1$. The spacings of the lines are $1/3$ unit. That the vectors pierce three of the lines representing their duals and do not pierce any of the lines representing the other basis elements is one way to see the relationship $\tilde{\mathbf{e}}^i(\mathbf{e}_j)(\mathbf{m}) = \delta^i_j$.